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Education

Seoul National University Ph.D. in Electrical and Computer Engineering.	Mar. 2013 - Aug. 2017
Seoul National University M.S. in Electrical and Computer Engineering.	Mar. 2011 - Feb. 2013
Seoul National University B.S. in Electrical Engineering and Computer Science.	Mar. 2006 - Aug. 2010

Publications

Junyoung Seo, Hyunwook Choi, Minkyung Kwon, Jinhyeok Choi, Siyoon Jin, Gayoung Lee, Junho Kim, JoungBin Lee, Geonmo Gu, Dongyoon Han, **Sangdoo Yun**, Seungryong Kim*, Jin-Hwa Kim*. "Grounding World Simulation Models in a Real-World Metropolis", *arXiv*, 2026.

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Ahmed Heakl, Martin Gubri, Salman Khan, **Sangdoo Yun***, Seong Joon Oh*. "Dr.LLM: Dynamic Layer Routing in LLMs", *ICLR*, 2026.

Junha Song, **Sangdoo Yun**, Dongyoon Han, Jaegul Choo, Byeongho Heo. "RL makes MLLMs see better than SFT", *ICLR*, 2026.

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Jang-Hyun Kim, Jinuk Kim, Sangwoo Kwon, Jae W. Lee, **Sangdoo Yun**, Hyun Oh Song. "KVzip: Query-Agnostic KV Cache Compression with Context Reconstruction", *NeurIPS* 2025.[Oral presentation]

Taekyung Kim, Dongyoon Han, Byeongho Heo, Jeongeun Park, **Sangdoo Yun**. "Token Bottleneck: One Token to Remember Dynamics", *NeurIPS* 2025.

Tommaso Green, Martin Gubri, Haritz Puerto, **Sangdoo Yun***, Seong Joon Oh*. "Leaky Thoughts: Large Reasoning Models Are Not Private Thinkers", *EMNLP* 2025.

Jeongeun Park, Sungjoon Choi*, **Sangdoo Yun***. "A Unified Framework for Motion Reasoning and Generation in Human Interaction", *ICCV* 2025.

Haneul Yoo, Cheonbok Park, **Sangdoo Yun**, Alice Oh, Hwaran Lee. "Code-Switching Curriculum Learning for Multilingual Transfer in LLMs", *ACL* 2025

Haritz Puerto, Martin Gubri, **Sangdoo Yun**, Seong Joon Oh. "Scaling Up Membership Inference: When and How Attacks Succeed on Large Language Models", *NAACL Findings* 2025.

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Changdae Oh, Yixuan Li, Kyungwoo Song*, **Sangdoo Yun***, Dongyoon Han*. "DaWin: Training-free Dynamic Weight Interpolation for Robust Adaptation", *International Conference on Learning Representations (ICLR 2025)*, 2025.

Yong-Hyun Park, **Sangdoo Yun**, Jin-Hwa Kim, Junho Kim, Geonhui Jang, Yonghyun Jeong, Junghyo Jo, Gayoung Lee. "Direct unlearning optimization for robust and safe text-to-image models", *Neural Information Processing Systems (NeurIPS 2024)*, 2024.

Changdae Oh, Hyesu Lim, Mijoo Kim, Dongyoon Han, **Sangdoo Yun**, Jaegul Choo, Alexander Hauptmann, Zhi-Qi Cheng, Kyungwoo Song. "Towards calibrated robust fine-tuning of vision-language models", *Neural Information Processing Systems (NeurIPS 2024)*, 2024.

Dong-Hwan Jang, **Sangdoo Yun***, Dongyoon Han*. "Model Stock: All we need is just a few fine-tuned models", *European Conference on Computer Vision (ECCV 2024)*, 2024. [Oral Presentation]

Wonjae Kim, Sanghyuk Chun, Taekyung Kim, Dongyoon Han, **Sangdoo Yun**. "HYPE: Hyperbolic Entailment Filtering for Underspecified Images and Texts", *European Conference on Computer Vision (ECCV 2024)*, 2024. [Oral Presentation]

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Geonmo Gu*, Sanghyuk Chun*, Wonjae Kim, Yoohoon Kang, **Sangdoo Yun**, "Language-only Efficient Training of Zero-shot Composed Image Retrieval", *IEEE Computer Vision and Pattern Recognition (CVPR 2024)*, 2024. (*Equal contribution)

Jang-Hyun Kim, Junyoung Yeom, **Sangdoo Yun***, Hyun Oh Song*, "Compressed Context Memory for Online Language Model Interaction", *International Conference on Learning Representations (ICLR 2024)*, 2024. (*Equal contribution)

Seungone Kim*, Jamin Shin*, Yejin Cho*, Joel Jang, Shayne Longpre, Hwaran Lee, **Sangdoo Yun**, Seongjin Shin, Sungdong Kim, James Thorne, Minjoon Seo, "Prometheus: Inducing Fine-grained Evaluation Capability in Language Models", *International Conference on Learning Representations (ICLR 2024)*, 2024. (*Equal contribution)

Geewook Kim, Hodong Lee, Daehee Kim, Haeji Jung, Sanghee Park, Yoonsik Kim, **Sangdoo Yun**, Taeho Kil, Bado Lee, Seunghyun Park, "Cream: Visually-Situated Natural Language Understanding with Contrastive Reading Model and Frozen Large Language Models", *Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, 2023.

Siwon Kim, **Sangdoo Yun**, Hwaran Lee, Martin Gubri, Sungroh Yoon, Seong Joon Oh, "ProPILE: Probing Privacy Leakage in Large Language Models", *Neural Information Processing Systems (NeurIPS 2023)*, 2023.

Jang-Hyun Kim, **Sangdoo Yun**, Hyun Oh Song, "Neural Relation Graph for Identifying Problematic Data", *Neural Information Processing Systems (NeurIPS 2023)*, 2023.

Song Park*, Sanghyuk Chun*, Byeongho Heo, Wonjae Kim, **Sangdoo Yun**, "SeiT: Storage-Efficient Vision Training with Tokens Using 1% of Pixel Storage", IEEE International Conference on Computer Vision (**ICCV 2023**), 2023. (*Equal contribution)

Dongyoon Han*, Junsuk Choe*, Dante Chun, John Joon Young Chung, Minsuk Chang, **Sangdoo Yun**, Jean Y. Song, Seong Joon Oh, "Neglected Free Lunch – Learning Image Classifiers Using Annotation Byproducts", IEEE International Conference on Computer Vision (**ICCV 2023**), 2023. (*Equal contribution)

Jaewoo Ahn, Yeda Song, **Sangdoo Yun**, Gunhee Kim, "MPChat: Towards Multimodal Persona-Grounded Conversation", **ACL 2023**.

Namuk Park, Wonjae Kim, Byeongho Heo, Taekyung Kim, **Sangdoo Yun**, "What Do Self-Supervised Vision Transformers Learn?", International Conference on Learning Representations (**ICLR 2023**), 2023.

Taeh Kim, Jinhyung Kim, Minh Shim, **Sangdoo Yun**, Myunggu Kang, Dongyoon Wee, Sangyoun Lee, "Exploring Temporally Dynamic Data Augmentation for Video Recognition", International Conference on Learning Representations (**ICLR 2023**), 2023.

Chanwoo Park*, **Sangdoo Yun***, Sanghyuk Chun, "A Unified Analysis of Mixed Sample Data Augmentation: A Loss Function Perspective", Neural Information Processing Systems (**NeurIPS 2022**), 2022. (*Equal contribution)

Geewook Kim, Teakgyu Hong, Moonbin Yim, Jeongyeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, **Sangdoo Yun**, Dongyoon Han, Seunghyun Park, "OCR-free Document Understanding Transformer", European Conference on Computer Vision (**ECCV 2022**), 2022.

Jang-Hyun Kim, Jinuk Kim, Seong Joon Oh, **Sangdoo Yun**, Hwanjun Song, Joonhyun Jeong, Jung-Woo Ha, Hyun Oh Song, "Dataset Condensation via Efficient Synthetic-Data Parameterization", International Conference on Machine Learning (**ICML 2022**), 2022.

Saehyung Lee, Sanghyuk Chun, Sangwon Jung, **Sangdoo Yun**, Sungroh Yoon, "Dataset Condensation with Contrastive Signals", International Conference on Machine Learning (**ICML 2022**), 2022.

Seulki Park, Youngkyu Hong, Byeongho Heo, **Sangdoo Yun**, Jin Young Choi, "The Majority Can Help The Minority: Context-rich Minority Oversampling for Long-tailed Classification", IEEE Computer Vision and Pattern Recognition (**CVPR 2022**), 2022.

Jungbeom Lee, Seong Joon Oh, **Sangdoo Yun**, Junsuk Choe, Eunji Kim, Sungroh Yoon, "Weakly Supervised Semantic Segmentation using Out-of-Distribution Data", IEEE Computer Vision and Pattern Recognition (**CVPR 2022**), 2022.

Jongin Lim, **Sangdoo Yun**, Seulki Park, Jin Young Choi, "Hypergraph-Induced Semantic Tuple Loss for Deep Metric Learning", IEEE Computer Vision and Pattern Recognition (**CVPR 2022**), 2022.

Byeongho Heo, **Sangdoo Yun**, Dongyoon Han, Sanghyuk Chun, Junsuk Choe, Seong Joon Oh, "Re-thinking spatial dimensions of vision transformers", IEEE International Conference on Computer Vision (**ICCV 2021**), 2021.

Jeesoo Kim, Junsuk Choe, **Sangdoo Yun**, Nojun Kwak, "Normalization Matters in Weakly Supervised Object Localization", IEEE International Conference on Computer Vision (**ICCV 2021**), 2021.

Sangdoo Yun, Seong Joon Oh, Byeongho Heo, Dongyoon Han, Junsuk Choe, Sanghyuk Chun, "Re-labeling ImageNet: from Single to Multi-Labels, from Global to Localized Labels", IEEE Computer Vision and Pattern Recognition (**CVPR 2021**), 2021.

Dongyoon Han, **Sangdoo Yun**, Byeongho Heo, YoungJoon Yoo, "Rethinking Channel Dimensions for Efficient Model Design", IEEE Computer Vision and Pattern Recognition (**CVPR 2021**), 2021.

Byeongho Heo*, Sanghyuk Chun*, Seong Joon Oh, Dongyoon Han, **Sangdoo Yun**, Youngjung Uh, Jung-Woo Ha. "AdamP: Slowing Down the Slowdown for Momentum Optimizers on Scale-invariant Weights", International Conference on Learning Representations (**ICLR 2021**), 2021. (*Equal contribution)

Sangdoo Yun, Seong Joon Oh, Byeongho Heo, Dongyoon Han, Jinhyung Kim, "VideoMix: Rethinking Data Augmentation for Video Classification", arXiv, 2020.

Hyojin Bahng, Sanghyuk Chun, **Sangdoo Yun**, Jaegul Choo, Seong Joon Oh, "Learning De-biased Representations with Biased Representations", International Conference on Machine Learning (**ICML 2020**), 2020.

Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, Youngjoon Yoo, "Cut-Mix: Regularization Strategy to Train Strong Classifiers with Localizable Features" IEEE International Conference on Computer Vision (**ICCV 2019**), 2019. [**Oral presentation**] (accept ratio=4.3%)

Jeonghun Baek, Geewook Kim, Junyeop Lee, Sungrae Park, Dongyoon Han, **Sangdoo Yun**, Seong Joon Oh, and Hwalsuk Lee, "What Is Wrong with Scene Text Recognition Model Comparisons? Dataset and Model Analysis" IEEE International Conference on Computer Vision (**ICCV 2019**), 2019. [**Oral presentation**] (accept ratio=4.3%)

Byeongho Heo, Jeessoo Kim, **Sangdoo Yun**, Hyojin Park, Nojun Kwak, and Jin Young Choi, "A Comprehensive Overhaul of Feature Distillation" IEEE International Conference on Computer Vision (**ICCV 2019**), 2019.

Sanghyuk Chun, Seong Joon Oh, **Sangdoo Yun**, Dongyoon Han, Junsuk Choe, and Youngjoon Yoo, "An Empirical Evaluation on Robustness and Uncertainty of Regularization Methods" Uncertainty & Robustness in Deep Learning, **ICML Workshop**, 2019.

Youngmin Baek, Bado Lee, Dongyoon Han, **Sangdoo Yun**, and Hwalsuk Lee, "Character Region Awareness for Text Detection", IEEE Computer Vision and Pattern Recognition (**CVPR 2019**), 2019.

Byeongho Heo, Minsik Lee, **Sangdoo Yun**, and Jin Young Choi, "Knowledge Distillation with Adversarial Samples Supporting Decision Boundary", Association for the Advancement of Artificial Intelligence (**AAAI 2019**), 2019

Byeongho Heo, Minsik Lee, **Sangdoo Yun**, and Jin Young Choi, "Knowledge Transfer via Distillation of Activation Boundaries Formed by Hidden Neurons", Association for the Advancement of Artificial Intelligence (**AAAI 2019**), 2019

Donghoon Lee, **Sangdoo Yun**, Sungjoon Choi, Hwiyeon Yoo, Ming-Hsuan Yang, and Songhwai Oh, "Unsupervised Holistic Image Generation from Key Local Patches", European Conference on Computer Vision (**ECCV 2018**), 2018.

Sangdoo Yun, Jongwon Choi, Youngjoon Yoo, Kimin Yun, and Jin Young Choi, "Action-Driven Visual Object Tracking with Deep Reinforcement Learning", IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**).

YoungJoon Yoo, Seonguk Park, Junyoung Choi, **Sangdoo Yun**, and Nojun Kwak, "Butterfly Effect: Bidirectional Control of Classification Performance by Small Additive Perturbation", arXiv:1711.09681, 2017.

Sangdoo Yun, Jongwon Choi, Youngjoon Yoo, Kimin Yun, and Jin Young Choi, "Action-Decision Networks for Visual Tracking with Deep Reinforcement Learning", IEEE Computer Vision and Pattern Recognition (**CVPR 2017**), 2017. (**Spotlight**)

Youngjoon Yoo, **Sangdoo Yun**, Hyung Jin Chang, Yiannis Demiris, and Jin Young Choi, "Variational Autoencoded Regression: High Dimensional Regression of Visual Data on Complex Manifold", IEEE Computer Vision and Pattern Recognition (**CVPR 2017**), 2017.

Jongwon Choi, Hyung Jin Chang, **Sangdoo Yun**, Tobias Fischer, Yiannis Demiris, and Jin Young Choi, "Attentional Correlation Filter Network for Adaptive Visual Tracking", IEEE Computer Vision and Pattern Recognition (**CVPR 2017**), 2017.

Junho Cho, **Sangdoo Yun**, Kyoungmu Lee, Jin Young Choi, "PaletteNet: Image Recolorization with Given Color Palette", 2nd **NTIRE: New Trends in Image Restoration and Enhancement** workshop and challenge on super-resolution in conjunction with (**CVPR 2017**), 2017.

Sangdoo Yun, Kimin Yun, Jongwon Choi, and Jin Young Choi, "Density-Aware Pedestrian Proposal Networks for Robust People Detection in Crowded Scenes", International Workshop on Crowd Understanding in conjunction with European Conference on Computer Vision (**ECCV 2016**), 2016.

Kimin Yun, Jongin Lim, **Sangdoo Yun**, Soo Wan Kim, and Jin Young Choi, "Attention-Inspired Moving Object Detection in Monocular Dashcam Videos", International Conference on Pattern Recognition (**ICPR 2016**), 2016.

YoungJoon Yoo, Kimin Yun, **Sangdoo Yun**, JongHee Hong, Hawook Jeong and Jin Young Choi, "Visual Path Prediction in Complex Scenes with Crowded Moving Objects", IEEE Computer Vision and Pattern Recognition (**CVPR 2016**), 2016.

Sangdoo Yun and Jin Young Choi, "Voting-based 3D Object Cuboid Detection Robust to Partial Occlusion from RGB-D Images", IEEE Winter Conference on Applications of Computer Vision (**WACV 2016**), 2016.

Hawook Jeong, **Sangdoo Yun**, Kwang Moo Yi, and Jin Young Choi, "Category Attentional Search for Fast Object Detection by Mimicking Human Visual Perception", IEEE Winter Conference on Applications of Computer Vision (**WACV 2015**), 2015.

Sangdoo Yun, Kimin Yun, Soo Wan Kim, Youngjoon Yoo, and Jiyeoup Jeong, "Visual Surveillance Briefing System: Event-based Video Retrieval and Summarization", IEEE International Conference on Advanced Video and Signal based Surveillance (**AVSS 2014**), 2014. (**Oral**)

Tushar Sandhan, Youngjoon Yoo, Hanjoo Yoo, **Sangdoo Yun**, and Moonsub Byeon, "Multi-Task Learning with Over-Sampled Time-Series Representation of a Trajectory for Traffic Motion Pattern Recognition", IEEE International Conference on Advanced Video and Signal based Surveillance (**AVSS 2014**), 2014.

Sangdoo Yun, Hawook Jeong, Woo-Sung Kang, Byeongho Heo, and Jin Young Choi, "Self-organizing Cascaded Structure of Deformable Part Models for Fast Object Detection", IEEE International Conference on Pattern Recognition (**ICPR 2014**), 2014.

Sangdoo Yun, Soo Wan Kim, Kwang Moo Yi, Haan-ju Yoo, Jin Young Choi, "Multiple ground plane estimation for 3D scene understanding using a monocular camera", Proc. 27th International Conference of Image and Vision Computing New Zealand (**IVCNZ 2012**), 2012. (**Oral**)

Professional Activities

Serve as an Area Chair for NeurIPS'24'25'26, CVPR'25,'26, ICLR'25'26, ECCV'24'26, ICCV'25, ICML'26
 Serve as a reviewer for CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, etc.

Patent

US Patent 14/532,483. "Method and apparatus for processing image", May, 2015, published.

Research Experience

Postdoctoral Researcher Perception and Intelligence Lab., Seoul National University	Sep. 2017 - Feb. 2018
Graduate Research Assistant Perception and Intelligence Lab., Seoul National University	Mar. 2011 - Aug. 2017
Visiting Student Robotics Lab., The University of Auckland, New Zealand.	Jan. 2016 - Jan. 2016

Awards and Honors

Ourstanding Reviewer Awards (**CVPR 2021**)
 Award (top 30%), Vision Meets Drones: A Challenge (**ECCV 2018 Workshop**)
 Best Paper Award, 30th Workshop on Image Processing and Image Understanding (**IPIU 2018**)
 Best Poster Award, 30th Workshop on Image Processing and Image Understanding (**IPIU 2018**)
 Best Paper Award, 29th Workshop on Image Processing and Image Understanding (**IPIU 2017**)

Work Experience

Development of Predictive Visual Intelligence Technology Apr. 2014 - present
 Sponsored by the ICT R&D program of MSIP/IITP. The goal of the project is developing predicting algorithms such as traffic flows.

Development of Real-time 3D Trajectory Estimation in Multi-camera Situation Jan. 2016 - Sep. 2016
 Sponsored by Samsung SDS. The goal of the project is developing 3D trajectory estimation algorithm using multi-camera.

UAV Visual Surveillance System Mar. 2014 - Feb. 2015
 Sponsored by Samsung S1. The goal of the project is developing visual surveillance algorithms for unmanned aerial videos such as moving object detection in a non-stationary camera.

Intelligent Visual Surveillance System Mar. 2010 - Aug. 2013
 Sponsored by Samsung Techwin Co., Ltd. The goal of the project is developing visual analysis algorithms, such as background subtraction, object tracking and behavior understanding, and integrated system.

Teaching Experience

Topics in Artificial Intelligence - Multimodal Deep Learning Theories and Applications Fall 2024
Lecturer

Special Topics in Artificial Intelligence - Deep Learning and Real-world Applications, Seoul National University Spring 2022
Lecturer

Convex Optimization, Seoul National University Fall 2013
Teaching Assistant for Prof. Jin Young Choi

Introduction to Random Variables and Random Processes, Seoul National University Spring 2013
Teaching Assistant for Prof. Jin Young Choi

Skills

Programming Languages:	Python, C++, C, MATLAB
Deep Learning Frameworks:	Tensorflow, PyTorch, Caffe, MatConvNet
Operating Systems:	Linux, Mac OS X, Windows

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